Applied Machine Learning Lab

B.Tech Semester 4

Experiment 10

EMG Gesture Classification using Machine Learning Models

Date: 15th April 2026

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AIM

To classify hand gestures using EMG signal data and evaluate multiple machine learning models for gesture recognition.

THEORY

Electromyography (EMG) signals measure electrical activity produced by muscles.
Signal Processing: EMG signals are time-series data
Feature Extraction: MAV and RMS
Models: Logistic Regression, KNN, Decision Tree, Random Forest, Extra Trees, SVM
Evaluation: Accuracy and F1-score
Confusion Matrix used for performance analysis

Commands/Code Used

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, ExtraTreesClassifier
from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, f1_score



# ----------------------------
# LOAD DATA (example CSV)
# ----------------------------
df = pd.read_csv("emg_data.csv")   # change path if needed

# ----------------------------
# FEATURE EXTRACTION (MAV & RMS)
# ----------------------------
```

```
def extract_features(data):
    features = []

    for i in range(len(data)):
        signal = data.iloc[i, :-1].values.astype(float)

        # MAV (Mean Absolute Value)
        mav = np.mean(np.abs(signal))

        # RMS (Root Mean Square)
        rms = np.sqrt(np.mean(signal**2))

        features.append([mav, rms])

    return np.array(features)

X_features = extract_features(df)

# Target
y = df.iloc[:, -1]   # last column as label

# ----------------------------
# TRAIN TEST SPLIT (70/30)
# ----------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X_features, y, test_size=0.3, random_state=42
```

```
)

# ----------------------------
# SCALING
# ----------------------------
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# ----------------------------
# MODELS
# ----------------------------
models = {
    "Logistic Regression": LogisticRegression(),
    "KNN": KNeighborsClassifier(),
    "Decision Tree": DecisionTreeClassifier(),
    "Random Forest": RandomForestClassifier(),
    "Extra Trees": ExtraTreesClassifier(),
    "SVM": SVC()
}

results = {}

# ----------------------------
# TRAIN & EVALUATE
# ----------------------------
```

```
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred, average='weighted')

    results[name] = [acc, f1]

    print(f"\n{name}")
    print("Accuracy:", acc)
    print("F1 Score:", f1)

# ----------------------------
# MODEL COMPARISON
# ----------------------------
results_df = pd.DataFrame(results, index=["Accuracy", "F1 Score"]).T

results_df.plot(kind="bar", figsize=(10,6))
plt.title("Model Performance Comparison")
plt.xticks(rotation=45)
plt.ylabel("Score")
plt.show()
```

RESULTS

Feature extraction applied on EMG signals.
Multiple models compared using accuracy and F1-score.
Model performance comparison graph generated.

CONCLUSION

EMG data was successfully used for gesture classification. Ensemble models like Random Forest and Extra Trees performed best.